

Physics 1101 Group Problem 1

Current theories indicate that dinosaurs and many other organisms became extinct after the Earth was struck by an asteroid. Such a collision will almost certainly happen again and, if it happens soon, could cause the extinction of the current dominant species on Earth, us. You have heard that there is an international program to scan space for this danger and wonder how large an asteroid they are looking for. By searching the internet, you find that about 5% of that asteroid's mass ended up as dust spread uniformly over Earth after eventually settling out of the upper atmosphere. Geological exploration indicates that, on the average, a layer of 0.020 grams per cm^2 of this dust, which is chemically different than the Earth's rock, covers the Earth's surface. Typical asteroids have a density of about 2.0 g/cm^3 .

Possibly Useful Information:

Earth's Radius = 6380 km

Distance from Earth to Sun = 150 million km

$$\text{Volume of sphere} = \frac{4}{3} \pi R^3$$

$$1 \text{ km} = 10^5 \text{ cm}$$

$$1 \text{ mile} \approx 1.6 \text{ km}$$

$$\text{Surface area of sphere} = 4 \pi R^2$$

$$\text{Volume density} = \frac{\text{mass}}{\text{volume}} \text{ or } \rho = \frac{M}{V}$$

$$\text{Surface density} = \frac{\text{Mass}}{\text{Surface Area}} \text{ or } \sigma = \frac{M}{A}$$

$$\text{Linear density} = \frac{\text{mass}}{\text{length}} \text{ or } \lambda = \frac{M}{L}$$